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(54) **2'-SUBSTITUTED-N6-SUBSTITUTED PURINE NUCLEOTIDES FOR RNA VIRUS TREATMENT**

2'-SUBSTITUIERTE-N6-SUBSTITUIERTE PURINNUKLEOTIDE ZUR BEHANDLUNG VON
RNA-VIREN

2'-SUBSTITUÉ-N6-SUBSTITUÉS NUCLÉOTIDES DE PURINE POUR LE TRAITEMENT DE VIRUS
À ARN

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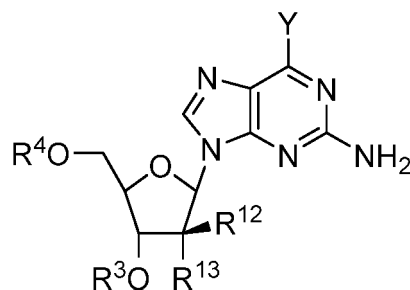
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hepacivirus or other than HCV with an effective amount of a compound of Formula I. Alternatively, a method is presented to treat a host, including a human, infected with a single-stranded RNA positive stranded virus of the *Flaviviridae* family, including but not limited to Dengue virus 2 and Yellow Fever, with an effective amount of a compound of Formula I. Alternatively, a method is presented to treat a host, including a human, infected with a single-stranded RNA negative-sense virus with an effective amount of a compound of Formula I. Alternatively, a method is presented to treat a host, including a human, infected with a double-stranded RNA virus with an effective amount of a compound of Formula I:



Formula I

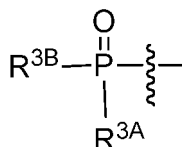
wherein:

Y is NR^1R^2 ;

R^1 is $\text{C}_1\text{-C}_5$ alkyl (including methyl, ethyl, n-propyl, iso-propyl, n-butyl, iso-butyl, sec-butyl, tert-butyl and pentyl), $\text{C}_1\text{-C}_5$ haloalkyl (including CH_2F , CHF_2 , CF_3 , CH_2CF_3 , CF_2CH_3 and CF_2CF_3), $\text{C}_2\text{-C}_6$ alkenyl, $\text{C}_2\text{-C}_6$ alkynyl, $-(\text{C}_0\text{-C}_2\text{alkyl})(\text{C}_3\text{-C}_6\text{cycloalkyl})$, $-(\text{Co-C}_2\text{alkyl})(\text{heterocycle})$, $-(\text{C}_0\text{-C}_2\text{alkyl})(\text{aryl})$, $-(\text{C}_0\text{-C}_2\text{alkyl})(\text{heteroaryl})$, $-\text{OR}^{25}$, $-\text{C}(\text{O})\text{R}^{3C}$ (including $-\text{C}(\text{O})\text{CH}_3$, $-\text{C}(\text{O})\text{CH}_2\text{CH}_3$, $-\text{C}(\text{O})\text{CH}(\text{CH}_3)_2$, $-\text{C}(\text{O})\text{OCH}_3$, $-\text{C}(\text{O})\text{OC}_2\text{H}_5$, $-\text{C}(\text{O})\text{OC}_3\text{H}_7$, $-\text{C}(\text{O})\text{OC}_4\text{H}_9$, and $-\text{C}(\text{O})\text{OC}_5\text{H}_{11}$), $-\text{C}(\text{S})\text{R}^{3D}$, or $-\text{SO}_2\text{R}^{28}$ each of which can be optionally substituted;

R^2 is hydrogen, $\text{C}_1\text{-C}_5$ alkyl (including methyl, ethyl, n-propyl, iso-propyl, n-butyl, iso-butyl, sec-butyl, tert-butyl and pentyl), $\text{C}_1\text{-C}_5$ haloalkyl (including CHF_2 , CHF_2 , CF_3 , CH_2CF_3 and CF_2CF_3), $-(\text{C}_0\text{-C}_2\text{alkyl})(\text{C}_3\text{-C}_6\text{cycloalkyl})$, $-\text{C}(\text{O})\text{R}^{3C}$ (including $-\text{C}(\text{O})\text{CH}_3$, $-\text{C}(\text{O})\text{CH}_2\text{CH}_3$, $-\text{C}(\text{O})\text{CH}(\text{CH}_3)_2$, $-\text{C}(\text{O})\text{OCH}_3$, $-\text{C}(\text{O})\text{OC}_2\text{H}_5$, $-\text{C}(\text{O})\text{OC}_3\text{H}_7$, $-\text{C}(\text{O})\text{OC}_4\text{H}_9$, and $-\text{C}(\text{O})\text{OC}_5\text{H}_{11}$), $-(\text{Co-C}_2\text{alkyl})(\text{aryl})$, $-(\text{C}_0\text{-C}_2\text{alkyl})(\text{heterocycle})$, $-(\text{C}_0\text{-C}_2\text{alkyl})(\text{heteroaryl})$; and wherein at least one of R^1 and R^2 is methyl, CH_2F , CHF_2 or CF_3 ;

R^3 is hydrogen,



, diphosphate, triphosphate, an optionally substituted carbonyl linked amino acid, or $-\text{C}(\text{O})\text{R}^{3C}$;

R^{3A} can be selected from O^- , OH, an $-\text{O}$ -optionally substituted aryl, an $-\text{O}$ -optionally substituted heteroaryl, or an optionally substituted heterocyclyl;

R^{3B} can be selected from O^- , OH, an optionally substituted N-linked amino acid or an optionally substituted N-linked amino acid ester;

R^{3C} is alkyl, alkenyl, alkynyl, $-(\text{C}_0\text{-C}_2)(\text{cycloalkyl})$, $-(\text{C}_0\text{-C}_2)(\text{heterocyclo})$, $-(\text{C}_0\text{-C}_2)(\text{aryl})$, $-(\text{C}_0\text{-C}_2)(\text{heteroaryl})$, $-\text{O}$ -alkyl, $-\text{O}$ -alkenyl, $-\text{O}$ -alkynyl, $-\text{O}-(\text{C}_0\text{-C}_2)(\text{cycloalkyl})$, $-\text{O}-(\text{C}_0\text{-C}_2)(\text{heterocyclo})$, $-\text{O}-(\text{C}_0\text{-C}_2)(\text{aryl})$, or $-\text{O}-(\text{C}_0\text{-C}_2)(\text{heteroaryl})$, each of which can be optionally substituted;

R^4 is a monophosphate, diphosphate, triphosphate, or a stabilized phosphate prodrug, including but not limited to a phosphoramidate, a thiophosphoramidate, or any other moiety that is metabolized to a monophosphate, diphosphate or triphosphate *in vivo* in the host human or animal; or

R^3 and R^4 together with the oxygens that they are bonded to can form a 3',5'-cyclic prodrug, including but not limited to, a 3',5'-cyclic phosphate prodrug;

R^{12} is hydrogen, CH_3 , CH_2F , CHF_2 , CF_3 , or ethynyl; and

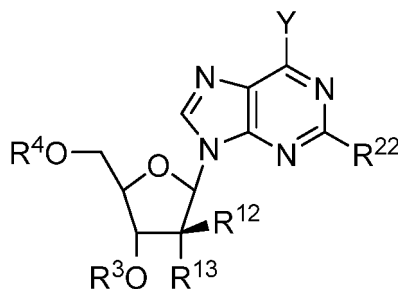
R^{13} is hydrogen, fluoro, chloro, bromo, N_3 , NH_2 , CN or OR^3 ;

wherein when R^{12} is methyl, R^{13} is bromo, chloro, N_3 , NH_2 , CN or OR^3 .

Alternatively, at least one of R^1 and R^2 is cyclopropyl.

Alternatively, at least one of R^1 and R^2 is cyclopentyl.

[0026] In another aspect described herein but not claimed, there is provided the use of a compound of Formula II below for the treatment of an infection of an RNA virus in a host, for example, a human in need thereof. A method is presented to treat a host, including a human, infected with a single-stranded RNA positive stranded virus other than a hepatitis virus or other than HCV with an effective amount of a compound of Formula II. Alternatively, a method is presented to treat a host, including a human, infected with a single-stranded RNA positive stranded virus of the *Flaviviridae* family, including but not limited to Dengue virus 2 and Yellow Fever, with an effective amount of a compound of Formula II. Alternatively, a method is presented to treat a host, including a human, infected with a single-stranded RNA negative-sense virus with an effective amount of a compound of Formula II. Alternatively, a method is presented to treat a host, including a human, infected with a double-stranded RNA virus with an effective amount of a compound of Formula II:



Formula II

wherein:

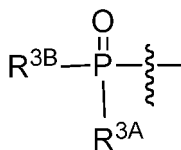
Y is NR¹R²;

R¹ is C₁-C₅alkyl (including methyl, ethyl, n-propyl, iso-propyl, n-butyl, iso-butyl, sec-butyl, tert-butyl and pentyl), C₁-C₅haloalkyl (including CH₂F, CHF₂, CF₃, CH₂CF₃, CF₂CH₃ and CF₂CF₃), C₂-C₆ alkenyl, C₂-C₆ alkynyl, -(C₀-C₂alkyl)(C₃-C₆cycloalkyl), -(C₀-C₂alkyl)(heterocycle), -(C₀-C₂alkyl)(aryl), -(C₀-C₂alkyl)(heteroaryl), -OR²⁵, -C(O)R^{3C} (including -C(O)CH₃, -C(O)CH₂CH₃-C(O)CH(CH₃)₂, -C(O)OCH₃, -C(O)OC₂H₅, -C(O)OC₃H₇, -C(O)OC₄H₉, and -C(O)OC₅H₁₁), -C(S)R^{3D}, or -SO₂R²⁸ each of which can be optionally substituted;

R² is hydrogen, optionally substituted C₁-C₅alkyl (including methyl, ethyl, n-propyl, iso-propyl, n-butyl, iso-butyl, sec-butyl, tert-butyl and pentyl), C₁-C₅haloalkyl (including CHF₂, CH₂F, CF₃, CH₂CF₃ and CF₂CF₃), optionally substituted -(C₀-C₂alkyl)(C₃-C₆cycloalkyl), optionally substituted -(C₀-C₂alkyl)(heterocycle), optionally substituted -(C₀-C₂alkyl)(aryl), optionally substituted -(C₀-C₂alkyl)(heteroaryl), -C(O)R^{3C} (including -C(O)CH₃, -C(O)CH₂CH₃-C(O)CH(CH₃)₂, -C(O)OCH₃, -C(O)OC₂H₅, -C(O)OC₃H₇, -C(O)OC₄H₉, and -C(O)OC₅H₁₁), -C(S)R^{3D}, or -SO₂R²⁸; and

wherein at least one of R¹ and R² is methyl, CH₂F, CHF₂ or CF₃;

R³ is hydrogen,



, diphosphate, triphosphate, an optionally substituted carbonyl linked amino acid, or -C(O)R^{3C};

R^{3A} can be selected from O⁻, OH, an -O-optionally substituted aryl, an -O-optionally substituted heteroaryl, or an optionally substituted heterocyclyl;

R^{3B} can be selected from O⁻, OH, an optionally substituted N-linked amino acid or an optionally substituted N-linked amino acid ester;

R^{3C} is alkyl, alkenyl, alkynyl, -(C₀-C₂)(cycloalkyl), -(C₀-C₂)(heterocyclo), -(C₀-C₂)(aryl), -(C₀-C₂)(heteroaryl), -O-alkyl, -O-alkenyl, -O-alkynyl, -O-(C₀-C₂)(cycloalkyl), -O-(C₀-C₂)(heterocyclo), -O-(C₀-C₂)(aryl), -O-(C₀-C₂)(heteroaryl), -S-alkyl, -S-alkenyl, -S-alkynyl, -S-(C₀-C₂)(cycloalkyl), -S-(C₀-C₂)(heterocyclo), -S-(C₀-C₂)(aryl), or -S-(C₀-C₂)(heteroaryl) each of which can be optionally substituted;

R^{3D} is alkyl, alkenyl, alkynyl, -(C₀-C₂)(cycloalkyl), -(C₀-C₂)(heterocyclo), -(C₀-C₂)(aryl), -(C₀-C₂)(heteroaryl), -O-alkyl, -O-alkenyl, -O-alkynyl, -O-(C₀-C₂)(cycloalkyl), -O-(C₀-C₂)(heterocyclo), -O-(C₀-C₂)(aryl), or -O-(C₀-C₂)(heteroaryl), each of which can be optionally substituted;

R⁴ is a monophosphate, diphosphate, triphosphate, or a stabilized phosphate prodrug, including but not limited to

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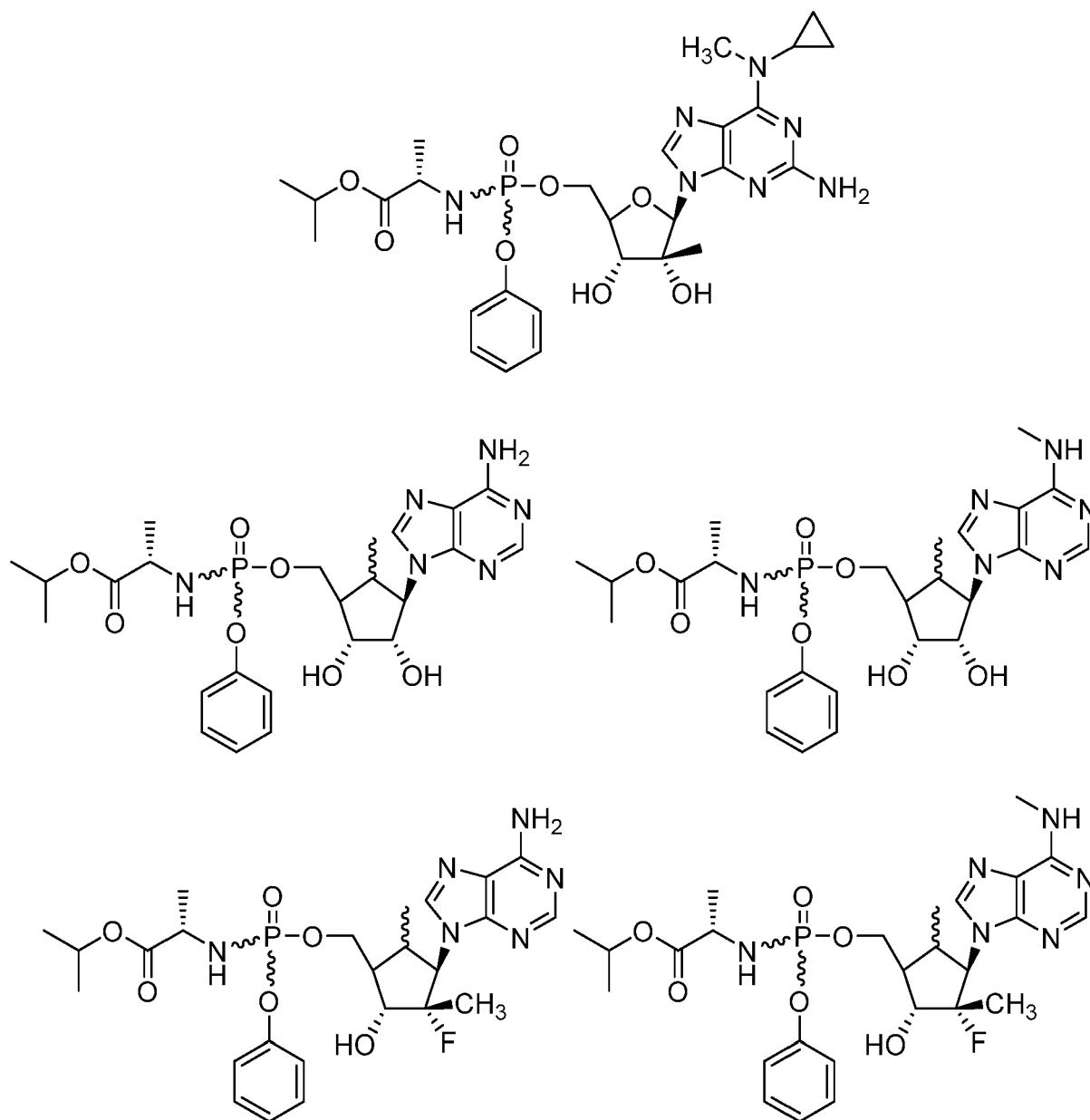
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[0076] Compounds and compositions for use in the treatment of a host infected with an RNA virus described herein, for example, other than a hepacivirus or HCV are provided. For example, the compounds for use of the invention can be administered in an effective amount alone or in combination with another anti-RNA viral agent to treat the infected host in need thereof. It may be useful to administer a combination of drugs that modulate the same or a different pathway or inhibit a different target in the virus. As the disclosed 2'-substituted-N⁶-substituted purine nucleotides are polymerase inhibitors, it can be advantageous to administer the compound to a host in combination with a protease inhibitor or an NS5A inhibitor. The compounds for use of the invention can also be administered in combination with a structurally different polymerase inhibitor such as another compound described herein or otherwise known to those in the art. The compounds for use of the invention can also be administered in combination with ribavirin and/or interferon.

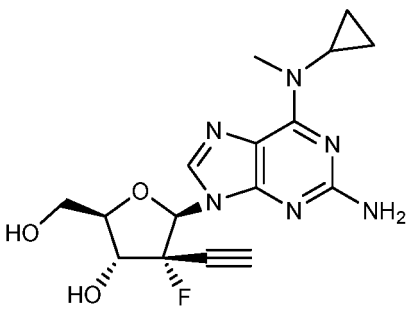
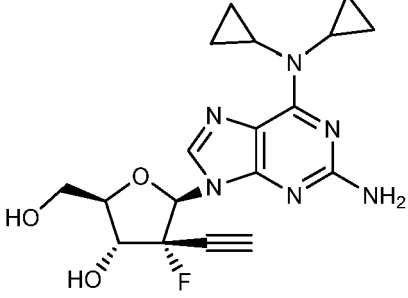
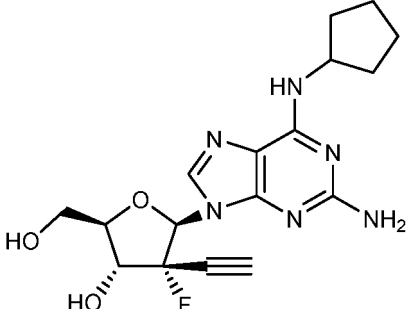
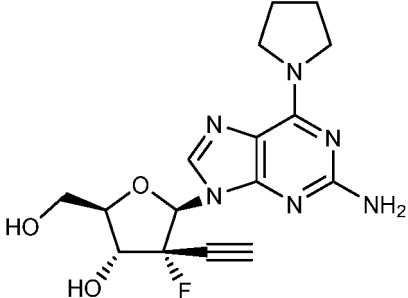
[0077] The 2'-substituted-N⁶-substituted purine nucleotides of the invention are typically administered orally, for example in pill or tablet form, but may be administered via another route which the attending physician considers appropriate, including via intravenous, transdermal, subcutaneous, topical, parenteral, or other suitable route.

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DETAILED DESCRIPTION OF THE INVENTION

[0078] The invention is according to the appended claims. Also disclosed herein but not claimed are compounds and their uses, methods, and compositions as described herein for the treatment of infections in or exposure to humans or

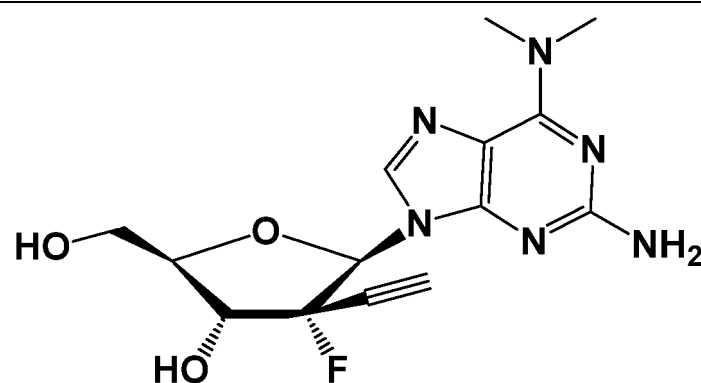
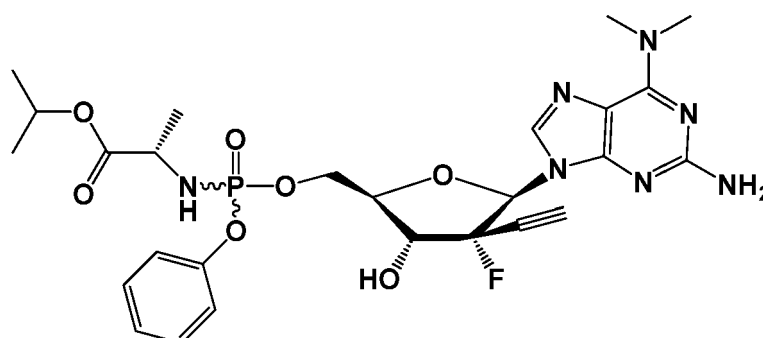
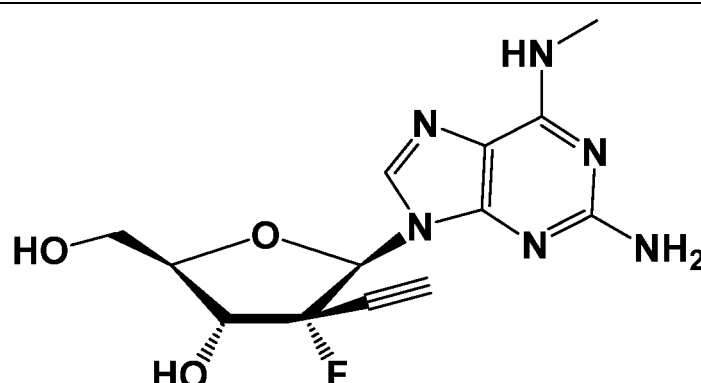
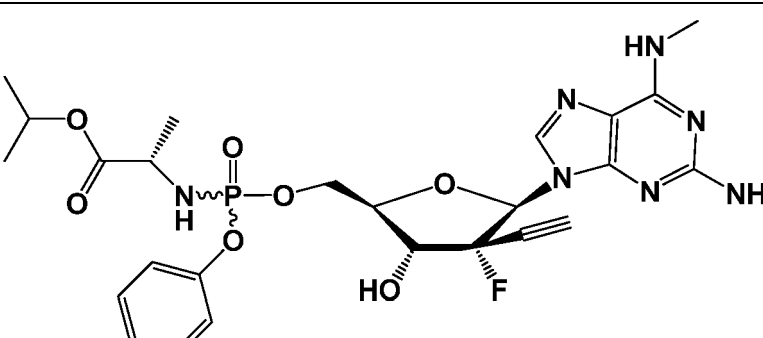
Table 2. Mass Spectra and ¹H NMR of Compounds 207, 208, 209, and 210 (not in accordance with the present invention)

Compd No.	Structure	¹ H NMR and Mass Spectra
207		¹ H NMR (400 MHz, Methanol-d ₄) δ 8.11 (s, 1H), 6.29 (d, J = 16.9 Hz, 1H), 4.76 (dd, J = 21.7, 9.0 Hz, 1H), 4.10 - 4.01 (m, 2H), 3.87 (dd, J = 13.1, 3.1 Hz, 1H), 3.37 (s, 1H), 3.24 - 3.11 (m, 2H), 1.00 - 0.87 (m, 2H), 0.74 (td, J = 4.6, 2.8 Hz, 2H). [M+H] ⁺ = 363.0
208		¹ H NMR (400 MHz, Methanol-d ₄) δ 8.16 (s, 1H), 6.30 (d, J = 16.8 Hz, 1H), 4.76 (dd, J = 21.6, 9.1 Hz, 1H), 4.10 - 4.01 (m, 2H), 3.87 (dd, J = 12.9, 2.9 Hz, 1H), 3.23 (s, 2H), 3.22 (s, 1H), 3.00 (tt, J = 7.0, 3.7 Hz, 2H), 1.21 (s, 4H), 1.03 - 0.86 (m, 4H), 0.77 (dt, J = 5.5, 3.4 Hz, 4H). [M+H] ⁺ = 389.0
209		¹ H NMR (400 MHz, Methanol-d ₄) δ 8.05 (s, 1H), 6.25 (d, J = 16.9 Hz, 1H), 4.77 (dd, J = 21.8, 9.2 Hz, 1H), 4.50 (s, 1H), 4.11 - 4.01 (m, 2H), 3.87 (dd, J = 13.0, 3.0 Hz, 1H), 3.37 (s, 2H), 3.21 (d, J = 5.4 Hz, 1H), 2.14 - 2.01 (m, 2H), 1.89 - 1.70 (m, 2H), 1.75 - 1.64 (m, 1H), 1.60 (s, 1H), 1.68 - 1.50 (m, 1H). [M+H] ⁺ = 377.0
210		¹ H NMR (400 MHz, Methanol-d ₄) δ 8.08 (s, 1H), 6.28 (d, J = 16.7 Hz, 1H), 4.76 (dd, J = 21.7, 9.0 Hz, 1H), 4.10 - 4.01 (m, 2H), 3.99 (s, 2H), 3.87 (dd, J = 13.1, 3.2 Hz, 1H), 3.37 (s, 1H), 3.19 (d, J = 5.4 Hz, 1H), 2.03 (s, 4H). [M+H] ⁺ = 363.1

Example 13. Non-limiting Examples of Compounds of the Present Invention

[0311] Compounds of the present invention include those in Table 3 and Table 4. Compound number, structure and molecular weight of illustrative compounds of the invention are also shown.

Table 3: Compounds of the Present Invention (* = not in accordance with the present invention)

Cmpd No.	Structure	MW
1*		336.3
2*		605.6
3*		322.3
4*		591.5